MARYAM SAIF

AI & Machine Learning Engineer | Data Scientist | Python Developer

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PROFESSIONAL SUMMARY

Highly motivated MS Computer Science graduate specializing in Data Science with strong expertise in Python development, Deep Learning, Explainable AI (XAI), and Medical Image Processing. Demonstrated ability to analyze complex data, build intelligent models, and develop efficient solutions. Research experience in Tri-Modal Malware Detection using CNN, Bi-LSTM, and TabNet with XAI techniques (SHAP, LIME, Grad-CAM). Eager to contribute to technology-driven organizations in Germany in AI/ML, Healthcare AI, and Data Science roles.

TECHNICAL SKILLS

Programming Languages:	Python, C++, C#, Java (Basics), HTML, CSS
AI / ML / Deep Learning:	NumPy, Pandas, Scikit-Learn, TensorFlow, PyTorch, OpenCV, MATLAB
Explainable AI (XAI):	SHAP, LIME, Grad-CAM, GNN Explainers, Model Interpretability
Core AI Skills:	Data Preprocessing, Model Training,Transfer Learning, Image Processing, Feature Engineering
Databases & Tools:	MySQL, SQLite, MS Office, Android Studio, VS Code, Jupyter Notebook, Google Colab, PyCharm
Soft Skills:	Research Writing, Problem Solving, Analytical Thinking, Team Collaboration

EDUCATION

MS Computer Science (Data Science) — CGPA: 3.25
University of Gujrat, Gujrat, Pakistan 2024 – 2026

BS Computer Science — CGPA: 3.39
University of Sialkot, Sialkot, Pakistan 2020 – 2024

Intermediate (ICS)
Punjab Group of Colleges Daska, Pakistan 2018 – 2020

Matriculation
Govt Girls Islamia High School Daska, Pakistan 2016 – 2018

PROJECTS

Chest X-Ray Disease Detection with Explainable AI (XAI)

PyTorch | OpenCV | Grad-CAM | ResNet18 | Transfer Learning | Python *Healthcare AI*

- Built a Deep Learning model to detect Pneumonia and COVID-19 from chest X-ray images using ResNet18 with Transfer Learning achieving 92%+ accuracy
- Implemented Grad-CAM heatmaps to visualize model focus areas — making AI transparent and interpretable for clinical use
- Dataset: Kaggle Chest X-Ray dataset (5,863 images) with Normal and Pneumonia classes
- Designed end-to-end pipeline: Image preprocessing → Model training → XAI visualization → Result output

Patient Readmission Prediction System

Python | Pandas | Scikit-Learn | XGBoost | Streamlit | Data Analytics *Data Science | Healthcare*

- Developed an ML model to predict 30-day hospital readmission risk using patient health records and clinical data
- Applied feature engineering, class imbalance handling (SMOTE), and hyperparameter tuning for optimal performance
- Built interactive dashboard using Streamlit for hospital staff to visualize risk scores and patient data
- Dataset: Kaggle Diabetes 130-US Hospitals dataset with 100,000+ patient records

PyTorch | CNN | Bi-LSTM | TabNet | SHAP | LIME | Grad-CAM | GMU Fusion *Research | XAI | Cybersecurity*

- Conducted multimodal malware classification research combining CNN (image features), Bi-LSTM (dynamic sequences), and TabNet (static features)
- Applied GMU-based multimodal fusion to integrate heterogeneous feature representations for accurate classification
- Implementing Explainable AI techniques — SHAP, LIME, Grad-CAM, GNN Explainers — for model interpretability
- Research demonstrates how XAI makes black-box security AI transparent for cybersecurity analysts

Cursor Control System via Hand Gesture Recognition

Python | OpenCV | MediaPipe | Image Processing | Real-time ML *Final Year Project | BS CS*

- Developed real-time cursor control system using hand gesture recognition to control mouse movements
- Enabled control of Media Player and PowerPoint slides using specific hand gestures
- Utilized image processing techniques and Python for gesture detection and model training
- Achieved low-latency real-time performance suitable for practical use

Food Donation Mobile Application

Android Studio | Java | Firebase | Mobile Development | UI/UX *Social Impact*

- Developed mobile application connecting food donors with recipients to minimize food wastage
- Designed user-friendly interface for easy food listing, search, and request management
- Implemented core functionalities: food posting, real-time updates, and user authentication via Firebase
- Promoted social responsibility and community engagement through technology

Medical AI Chatbot for Symptom Analysis

Python | HuggingFace Transformers | NLP | Streamlit | Medical NLP *NLP | Healthcare AI*

- Built AI-powered chatbot that analyzes patient symptoms and suggests possible conditions with confidence scores
- Used HuggingFace pre-trained medical NLP models for intent recognition and symptom extraction
- Designed Streamlit-based web interface with clean UI for patient and doctor interaction
- Includes disclaimer and recommendation to consult a professional for medical advice

RESEARCH EXPERIENCE

Tri-Modal Malware Detection

Independent Research — University of Gujrat 2024 – Present

- Performed multimodal malware detection using CNN for image features, Bi-LSTM for dynamic behavioral sequences, and TabNet for static analysis features
- Applied GMU-based multimodal fusion to integrate heterogeneous feature representations for accurate malware classification
- Currently implementing Explainable AI (SHAP, LIME, Grad-CAM, GNN Explainers) to make model decisions transparent and interpretable

LANGUAGES

English — Professional Working Proficiency

Urdu — Native / Full Professional Proficiency